ANSWER KEY & MARKING SCHEME SET A

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 49	Add the values, subtracting a smaller symbol that comes before a larger one
2	**(a)** 1994	Add the values, subtracting a smaller symbol that comes before a larger one
3	**(c)** XCIX	99 = XCIX
4	**(b)** 11110	Repeated division of 30 by 2
5	**(a)** 112	Repeated division of 44 by 6
6	**(b)** 23	Add digit × place value in base 3
7	**(b)** India	Hindu numerals
8	**(a)** Fibonacci	Liber Abaci
9	**(d)** subtracted	IV = 4
10	**(b)** 4 hundreds, i.e. 400	Place value

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Pairing each object with exactly one other object (1); each cow paired with one stick (1).	1+1
12	MMXXIV (2).	2
13	The number of distinct digits used, and the number each place is multiplied by (2).	2
14	No zero or place value, long strings of symbols (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	1444 = MCDXLIV (2); grouped by thousands, hundreds, tens and ones (1).	2+1
16	399 = CCCXCIX (2); grouped by thousands, hundreds, tens and ones (1).	2+1
17	2499 = MMCDXCIX (2); grouped by thousands, hundreds, tens and ones (1).	2+1
	OR: 1666 = MDCLXVI (2); grouped by thousands, hundreds, tens and ones (1).	2+1
18	3888 = MMMDCCCLXXXVIII (2); grouped by thousands, hundreds, tens and ones (1).	2+1
	OR: 949 = CMXLIX (2); grouped by thousands, hundreds, tens and ones (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) To track livestock, trade and time (2). (ii) One-to-one mapping (2). (iii) Cannot count beyond the number of letters (1).	2+2+1
	OR: (i) MCMXCIV (2.5). (ii) 1944 (2.5).	2½+2½
20	(i) 58 (2). (ii) 111010_2 (2). (iii) $10001_2 = 17$ (1).	2+2+1
	OR: (i) $3 \times 3600 = 10800$ s (1). (ii) 60 has many divisors: 2, 3, 4, 5, 6, 10, 12, 15, 20, 30 (2). (iii) 205 min (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

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ANSWER KEY & MARKING SCHEME SET B

Chapter 3: **A Story of Numbers** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 94	Add the values, subtracting a smaller symbol that comes before a larger one
2	**(b)** 2024	Add the values, subtracting a smaller symbol that comes before a larger one
3	**(c)** CD	400 = CD
4	**(a)** 122	Repeated division of 17 by 3
5	**(a)** 202	Repeated division of 100 by 7
6	**(a)** 64	Add digit × place value in base 4
7	**(d)** Bakhshali manuscript	c. 3rd century CE
8	**(b)** 60	Sexagesimal
9	**(b)** one-to-one mapping	Basic mechanism of counting
10	**(b)** zero	No zero and no place value

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$8+0+2+1=11$ (2).	2
12	$19=10011_2$ (2).	2
13	Ten symbols give every number using place value and zero, making arithmetic simple (2).	2
14	Egyptian, Mesopotamian (also Roman, Maya) (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Repeated division by 2 (2); 45 = 101101_2 (1).	2+1
16	Repeated division by 3 (2); 60 = 2020_3 (1).	2+1
17	Repeated division by 5 (2); 88 = 323_5 (1).	2+1
	OR: Repeated division by 6 (2); 123 = 323_6 (1).	2+1
18	Repeated division by 8 (2); 200 = 310_8 (1).	2+1
	OR: Repeated division by 4 (2); 77 = 1031_4 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Aryabhata explained the ten-symbol system fully; Al-Khwarizmi spread it in the Arab world (3). (ii) Originated in India, transmitted through the Arab world (2).	3+2
	OR: (i) $2\times10^3+3\times10^2+0\times10+4$ (2). (ii) $2\times125+3\times25+0+4=329$ (3).	2+3
20	(i) $2^8=256$ (2). (ii) 255 (1). (iii) 11001000_2 (2).	2+1+2
	OR: (i) $100-47=53$ (3). (ii) LIII (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.